

Data careers skill

The top skills needed for a data career

- All data professionals share a love of data and a desire to solve problems. While wearing their analytics hat, data professionals lay out the story that they're tempted to tell. Then they poke it from several angles with follow-up investigations to see if it holds water before bringing it to their decision-makers. In doing so, they rely on their programming and investigative skills to guide others towards informed decisions
- Data professionals also combine a knowledge about how to do practical tasks with an awareness of what makes communication and collaboration successful.

Important skills

Interpersonal skills

- Often, these are referred to as people's skills. They focus on communicating and building relationships.
- Interpersonal skills are critical

Active listening

- Allowing team members, bosses, and other collaborative stakeholders to share their own points of view before offering responses

As a data professional, you'll search for information hidden within a large amounts of data by applying critical thinking skills.

you'll investigate the connections between a variety of different data sources, as you search for trends and indicators. Think of yourself as a data detective.

Data cleaning

- This is where the data is reorganized and reformatted. The goal is to remove anything that could create an error during analysis.
- This process includes tagging and consolidating duplicates, irrelevant entries, structural errors, and empty space. Once you have everything in

the proper format, you can then filter out unwanted material

Ideal qualities for data analytics professionals

You have been learning about skills that can help you enter and excel in the data career space. In this program, you will be building technical abilities, which are necessary before pursuing opportunities in data analytics. Job postings will include a list of the required technical skills prominently displayed. Quite often, organizations will also make note of additional skills and traits that go beyond working with data on a computer. In this reading, you will explore examples of additional skills and traits that employers are seeking when searching for data analytics professionals.

As you begin to search for job opportunities, many employers seek additional skills that are not exclusive to digital fields. In the sections below, you will learn more about these traits through excerpts found within data analytics job postings.

Being coachable

Coachable individuals are capable of receiving feedback and using that information to make improvements. At the center of being coachable is a positive attitude, and the ability to self-reflect and take steps to grow. People who are coachable usually have a growth mindset, which is a belief that hard work and determination can make them better. As a result, they view feedback from colleagues and supervisors as an opportunity to improve their skill set.

A passion for data analysis

Employers often seek candidates whose commitment to data analysis extends beyond their professional duties. Volunteering your data skills to help a nonprofit organization is just one example, but it's not the only way to show your commitment to data analytics. Data analysis is applicable outside of the workplace, but is often not obvious. Community projects, helping a local school organize data, and developing your own side project are a few examples of how you might demonstrate your passion for data analysis outside of the workplace.

Another way to explore your passion for data analysis is to connect with other passionate data analysts and take on data challenges. A great website for

exploring data analytics is [Kaggle.com](https://www.kaggle.com), which hosts an active online community for data scientists and machine learning enthusiasts. Users can collaborate with other users, publish datasets, use GPU-integrated notebooks, and compete with other data scientists to solve data science challenges. Participating in activities hosted by an online data science community like Kaggle can add fuel to your passion for data analysis and provide artifacts for your portfolio.

Employers are seeking passionate candidates. Job postings that identify candidates that have a passion for data analysis often include language like:

- Seeking a passionate data scientist.
- We seek a candidate with deep curiosity directed toward diverse research interests.
- Driven. The prospect of focusing on corporate environmental impact as a data domain excites you, and you are personally motivated.

Pro tip: Include in your portfolio your passion for data analysis. Provide examples of passion projects, volunteer work, or analysis outside of employment to relay your commitment to data analysis. If you are conducting data analysis in your free time, it says a lot about your passion.

Lifelong learning

Earlier, you read about some generational shifts in data workspaces. You may recall that these shifts have included technological advances in computer interfaces, data storage, and the role of data analysis in organizations. In fact, the one consistent element through all of these eras was change. As you progress in your career, you will continue to learn new techniques, tools, and ways of interacting with other professionals. As you have learned, staying up to date with the latest technologies and techniques is essential for data professionals. Here are samples from actual job posting:

- Candidates must balance their hands-on work with a desire to keep up with trends.
- Seeking candidates with the aptitude and enthusiasm to develop new skills and areas of expertise.
- Wanting those with a drive to learn and master new technologies and techniques.

Pro tip: Stay current by reading data-related blogs and attending workshops. Visit business networking websites to connect and learn from subject matter experts.

Strong interpersonal skills

You may recall that data professionals and business intelligence professionals interact closely; together, they influence an organization's decision-making. Throughout a project, a data analyst will interact with many stakeholders, from different areas of an organization. Often, these interactions will extend to include clients, users, or representatives from other companies. Additionally, data professionals will work closely with a variety of other professionals as a member of a cross-functional team. Examples from job postings include interpersonal skill requirements for candidates and will include these types of statements:

- We seek employees that can build relationships internally to transfer knowledge, consult with fellow data scientists and analysts to guide analysis, and deliver larger projects.
- Wanting to hire those with the ability to negotiate complex and/or sensitive issues; and maintain good working relations.
- Our data professionals value building strong relationships with colleagues and partners.

Pro tip: Include interpersonal skills on your list of qualifications on your resume. Revisit the material within this program (including the material on communication you will be introduced to later in this course). Find opportunities to add examples of teamwork, empathy, leadership, mediating, and active listening.

Communication

Professionals working in data, business, and technology are expected to be fluent communicators. The effectiveness of your work will weigh heavily on your ability to inspire people through the results of your analysis. Job postings can contain requirements like the following:

- Candidates must have excellent communication skills and a friendly, approachable personality.

- You're a compelling storyteller who can communicate in succinct and inspiring ways to audiences with varied data science experience to influence real world product or feature decisions.
- A storyteller. You know that no data speaks for itself and take pride in the visualizations and narrative that you construct to communicate your careful analysis.

Pro tip: Be sure that all correspondences you exchange are professional and free of grammatical and spelling errors. Include any examples of written communication in your portfolio—these may include but are not limited to reflection pieces, executive summaries, or project proposals. Don't forget to include writing examples or online blog entries that describe how you have communicated in past situations.

Problem solver

The ability to resolve problems is an important part of being a data analytics professional, whether it's dealing with incomplete data, resolving issues within an analysis, or finding the best way to communicate your results to your audience. It is important that you also quantify the results of problems that you have solved. Organizations who seek data professionals have included problem-solving in their job descriptions, such as:

- Seeking data professionals who have the capability and the strong desire to solve problems from concept development, customer engagement, and technology transition.
- Candidates must be persistent and have excellent analytical and problem-solving skills.
- A big thinker. You start with "why" when approaching a new problem, and are always wanting to generalize, synthesize, and summarize.

Pro tip: Adjust your past working responsibilities into tasks that you were able to achieve. This can be accomplished by stressing the end results of your actions. Detail how your action or task had a direct impact on the organization. If there is a measured or empirical amount of change associated with your actions, be sure to include that. Example: Analyzed data from over 3,000 users to optimize systems, which led to a 32% increase in customer satisfaction.

Key takeaways

Employers in a variety of industries are eager to hire data professionals who possess more than just technical expertise. You can highlight your non-technical abilities for potential employers in several ways. Discover ways to successfully and effectively communicate your passions. By demonstrating a commitment to professional growth and sharing examples of your ability to solve problems, you can set yourself apart from other candidates.

Volunteer data skills to make a positive impact

So far, you have been learning how data analytics professionals are making an impact in a variety of industries. You've also learned about nonprofit organizations, and how their efforts are making a difference for so many in need. By volunteering their expertise and time, data analytics professionals contribute to many projects that help nonprofits benefit communities all around the world. In this reading, you will learn more about how you can contribute to global efforts, including your local community as a data professional.

Nonprofits

People hoping to further a social cause, or provide a benefit to the public, often create nonprofit groups. The main mission of a nonprofit group is to fill a need or gap that the private sector is not addressing adequately. There are rewarding and inspiring opportunities for data professionals in the nonprofit sector. At the global level, groups like [Children International](#), [Doctors Without Borders](#), and [Care International](#) lead humanitarian efforts across the globe. In addition to these large-scale efforts, there are many nonprofit organizations whose efforts are regionally specific, like [Christopher's Promise](#) based in the Columbus, Ohio area.

You can apply your data skills to help nonprofit groups become more effective, allowing them to anticipate and respond to their greatest areas of need.

You may already be aware of nonprofits that fit within your areas of interest or operate in your community. One way that these organizations compensate for their lack of resources is by finding volunteers. Often, regional or local organizations lack the resources to offer full time employment for marketing professionals or data analysts. As a data analytics professional there are many ways that you can make a valuable contribution to nonprofit organizations. In the event that you do not have a specific nonprofit organization in mind, you

can research through [Charity Navigator](#), the largest and most trusted online nonprofit evaluation and rating tool in the world. Here you will find ratings on nearly 200,000 nonprofits.

An example of a community-based nonprofit organization

[Christopher's Promise](#), a grass-roots nonprofit based in Columbus, Ohio, is a representative example of the types of community-based nonprofits that are making an impact all over the world. This organization was inspired by the efforts to create a bicycle for a fifteen-year old who faced the challenges of Neurofibromatosis, Cerebral Palsy, and Optic Glioma. Since 2011, this nonprofit has facilitated the design and construction of adaptive bicycles for hundreds of children, thanks to outstanding support from the local communities in the central Ohio area.

Christopher's Promise (2022) mission statement is "To allow all kids, despite physical limitations, the ability to experience the same hallmark childhood memories as their peers. Helping kids, be kids."

Volunteering data skills to nonprofits

As a data analytics professional, your skills can make a great impact on charity and nonprofit organizations within your own community or for a community whose needs resonate with you. Many nonprofits are highly dependent on volunteers for their activities due to limited resources. This situation is especially challenging for community-based initiatives that often lack the funding to hire data analytics professionals. Therefore, offering to volunteer with a local nonprofit is an excellent way to gain experience and support a cause.

How can data analytics professionals help nonprofits?

Data analytics professionals can help nonprofit organizations in a variety of ways. Below are a few examples.

Fundraising

Using data analytics, you can predict with high accuracy which prospects are most likely to donate. In order to maximize return on marketing and outreach

expenses, nonprofit organizations can use data analysis to prioritize outreach and reduce overall marketing expenditures.

Marketing

There is valuable information in a nonprofit's data that can benefit marketing efforts. Data analysis can help determine a nonprofit's target audience and gather insights such as donation methods, contribution history, and other demographics. As a result, the nonprofit can help guide communication with potential donors and determine the effectiveness of campaigns.

Monitoring activities

Data analysis can help nonprofits monitor resources, expenses, and daily operational needs. A nonprofit will often need assistance in keeping track of donated materials. For example, a database of donations to a food bank could help the organization determine specific food items being collected and inventory the expiration dates of food items for shelving, cycling, and distribution. Using a system of real-time inventory in conjunction with predictive data analytics could anticipate shortages and identify those donors who have helped provide these items in the past.

Data analytics has a lot to offer a nonprofit organization. Gaining deeper insight into the organization's efforts can help them to operate in a more effective and efficient way. Analyzing data can improve various nonprofit functions, such as allocation of resources, recruiting and retaining supporters, fundraising, and conducting research. Data analysis also can help reveal patterns and power dynamics that can be used to inform decision-making. In addition to identifying individuals and communities in need of assistance, predictive modeling can also be used to identify potential recipients of a nonprofit's services.

More opportunities to volunteer

There are organizations who are helping to pair data professionals with projects that are making a difference all around the world.

DataKind is a global nonprofit that harnesses the power of data science and artificial intelligence ("AI") in the service of humanity. Using data analysis, this organization guides restorative efforts in underserved communities. DataKind brings together volunteer experts from academia and industry to design innovative solutions to tough social challenges. They help social organizations apply their data to predictive analytics, machine learning algorithms, and AI in a

way that both increases impact and is sensitive to ethical considerations. Visit the [DataKind website](#) to find out more about their mission and how to become a volunteer.

Another option for putting your data skills to good use are hackathons. A hackathon is an event where programmers and other data professionals come together and collaborate on a particular project. The goal is to create a solution to an existing problem.

There are many hackathon events scheduled across the world each year. These events can take place at a specific location or remotely across the globe. Hackathon events are commonly built around a central theme or question, such as tracking industrial emissions, promoting healthy food options, or how we can improve mental health for cancer patients. If you are interested in finding listings of current hackathons around the world, visit [Devpost.com](#).

Key takeaways

Donating your time to a nonprofit is a great way to gain experience, and it can be mutually beneficial for you and the nonprofit. The nonprofit benefits from your data analysis expertise, and you get valuable experience working with data that can be added to your professional portfolio.

Working with a nonprofit organization or participating in a hackathon event introduces you to different types of projects, enhances interpersonal and communications skills, and adds to your professional list of contacts. No matter what your level of data analytics experience, working with nonprofits and hackathons can be a rewarding side project, or even a philanthropic career option.

Resources for more information

- [Charity Navigator](#) is the world's largest and most trusted nonprofit evaluator.
- [DataKind](#) helps social organizations identify their data and artificial intelligence opportunities, recruits and manages volunteers, and then sees the solutions are used.
- [Devpost](#) is a place to build products, practice skills, learn technologies, sign up for competitions, and grow your network.

- [IRS.gov tax exempt organizations listing](#) is a website to find charitable organizations throughout the United States.
- [Statistics without borders](#) contributes to the common good by providing free statistical, data science, and analytical services.

Important ethical considerations for data professionals

Personally Identifiable Information (PII)

- PII permits the identity of an individual to be inferred by either direct or indirect means. This includes things like biometric records, usernames, and Social Security or national identification numbers. Because this information is often associated with medical, financial, and employment records, PII is sensitive and must be managed with great care.
- Information that permits the identity of an individual to be inferred by either direct or indirect means
- Example:
 - Biometric records
 - Usernames
 - Social security or national identification numbers

Aggregate information

- Data from a significant number of users that has eliminated personal information
- By aggregating the data and removing PII, this protects people and gives them more control over their own data.

A person's background, experiences, and beliefs lead to bias, which may negatively affect data work.

A key thing to keep in mind is that data gathering is a task managed by humans, and that process can be informed by different backgrounds, experiences, beliefs, and worldviews. These and other types of biases can affect the way that data is communicated and how the results are shared, which in turn can have an impact on business decisions

Effective data professionals know that, whether collecting, analyzing, interpreting, or communicating sensitive data, bias should always be considered. So be very careful when interpreting data where there is a clear source of bias and be on the lookout for subtle biases as well.

In addition to thinking through bias in the data, data professionals should also try to emphasize that there can be a multitude of possible interpretations for every data insight.

The main trick is avoid jumping to conclusions until you've really done your homework.

Critical data security and privacy principles

You have learned how data analytics can be used for good causes, like assisting nonprofit organizations. Also, you learned that data professionals need to protect privacy within data and remain aware of other considerations, like data bias and making assumptions about data.

As a data analytics professional, you have a responsibility to handle data ethically. Data ethics refers to well-founded standards of right and wrong that dictate how data is collected, shared, and used. Throughout your career you will work with a lot of data. This sometimes includes PII, or **personally identifiable information**, which can be used by itself or with other data to track down a person's identity. One element of treating data ethically is ensuring that the privacy and security of that data is maintained throughout its lifetime. In this reading, you will learn more about the importance of data privacy and some strategies for protecting the privacy of data subjects.

Privacy matters

Data privacy means preserving a data subject's information and activity any time a data transaction occurs. This is also called information privacy or data protection. Data privacy is concerned with the access, use, and collection of personal data. For the people whose data is being collected, this means they have the right to:

- Protection from unauthorized access to their private data
- Freedom from inappropriate use of their data

- The right to inspect, update, or correct their data
- Ability to give consent to data collection
- Legal right to access the data

In order to maintain these rights, businesses and organizations have to put privacy measures in place to protect individuals' data. This is also a matter of trust. The public's ability to trust companies with personal data is important. It's what makes people want to use a company's product, share their information, and more.

Protecting privacy with data anonymization



Organizations use a lot of different measures to protect the privacy of their data subjects, like incorporating access permissions to ensure that only the people who are supposed to access that information can do so. Another key strategy to maintaining privacy is data anonymization.

Data anonymization is the process of protecting people's private or sensitive data by eliminating PII. Typically, data anonymization involves blanking, hashing, or masking personal information, often by using fixed-length codes to represent data columns, or hiding data with altered values.

Data professionals can take additional measures to protect users and their data.

Data aggregation, for example, is the process of collecting and combining details from a significant number of users in terms of totals or summary.

Aggregating data ensures that information contained within datasets is shown in groups; when coupled with other anonymization techniques, data professionals can ensure compliance with data privacy and anonymization standards.

Data anonymization is used in just about every industry. As a data analytics professional, you probably won't personally be performing anonymization, but it's useful to understand what kinds of data are often anonymized before you start working with it. This data might include:

- Telephone numbers
- Names
- License plates and license numbers

- Social security numbers
- IP addresses
- Medical records
- Email addresses
- Photographs
- Account numbers

Imagine a world where we all had access to each other's addresses, account numbers, and other identifiable information. That would invade a lot of people's privacy and make the world less safe. Data anonymization is one of the ways we can help keep data private and secure!

Key takeaways

For any professional working with data about actual people, it's important to consider the safety and privacy of those individuals. That's why understanding the importance of data privacy and how data that contains PII can be made secure for analysis is so important. We have a responsibility to protect people's data and the personal information that data might contain.

If you're interested in learning more about data privacy and ethics, you can check out [the Google Data Analytics Certificate program's section on bias, credibility, privacy, ethics, and access](#)

The practices and principles of good data stewardship

As you have been learning, all data professionals are responsible for ensuring the quality, integrity, accessibility, and security of data. Data stewardship is the practice of ensuring that data is accessible, usable, and safe. Making data stewardship a normal part of your work habits will benefit everyone who relies on your analysis, both inside and outside of your organization. In this reading, you will learn more about data stewardship and receive some best practices that can assist in guiding your career in data analytics.

Respect privacy

Earlier in this course, you learned about Information that permits the identity of an individual to be inferred by either direct or indirect means. This kind of information is commonly referred to as personally identifiable information or PII. When users share personal information, they are putting a high level of trust into an organization. It is the responsibility of all who have access within the organization to help protect the privacy of their users. As a data analytics professional, it is important to be thoughtful about any personal data and exhibit great care to protect it. In different parts of the world, laws are in place to guide best practices for data privacy. Laws provide a foundation for best practices as you grow in knowledge and experience on how to support and sustain privacy. One of your responsibilities as a data professional will be to stay up to date with any change in data laws and regulations that govern data. Depending on your organization's location or industry considerations, there may be additional regulations and policies in place. Here are a couple of regional examples:

- General Data Protection Regulation or GDPR (European Union law):
 - The GDPR is described on their website as the toughest privacy and security law in the world. It imposes obligations onto organizations anywhere, so long as they target or collect data related to people in the European Union.
- Lei Geral de Proteção de Dados Pessoais or LGPD (Brazil's general law for the protection of personal data):
 - The LGPD is a data protection law that governs how companies collect, use, disclose, and process personal data belonging to people in Brazil. LGPD applies to companies that process data about individuals in Brazil.
- The California Consumers Privacy Act or CCPA (Privacy rights for California consumers):
 - The CCPA gives consumers more control over the personal information that businesses collect about them. These regulations provide guidance on how to implement the law.
 - Additionally, states like Virginia, Colorado, New York, Utah, and Connecticut have enacted similar legislation to protect consumer privacy in their states.

Be cautious of unintentional harm

Data analytics is expanding its influence across an increasing range of industries. Companies are using the results of data analysis to make informed decisions. Many of these decisions have the potential to impact people across a broad range of social and economic factors. It is good practice to continually strive to produce information that is accurate, while respecting cultural and social norms.

Due to the global marketplace, decisions play out differently in different cultures. Taking these issues and considerations into account is very important for the executive team of an organization. Also, companies are known to take a position on particular politicized social and cultural issues, and these can be reflected in their policies. As a data analytics professional, you must be cognizant of your company's policies. When presented with challenges, it is best to seek guidance from leadership within your organization on how to navigate.

Avoid creating or reinforcing bias

You have learned about bias within data and how it can have an impact on your analysis. Identifying bias is not always simple. A good practice when working with data is to keep in mind that data gathering is a task managed by humans—and that process is informed by people from different backgrounds, experiences, beliefs, and worldviews. These and other types of biases can affect the data and the results, which in turn can have an impact on business decisions. You will learn more about bias within data as you progress through the program.

Consider inclusivity

Often in your role as a data analytics professional, you will have access to data collected in a variety of ways. You will need to consider whether the methods of data collection have excluded information from particular populations. Inclusionary approaches can expand how any organization collects and analyzes data. Building diverse research teams, communicating clearly with user communities, and engaging in careful and critical analysis that considers equity and inclusion benefits all stakeholders.

Uphold high standards of scientific excellence

The processes and technology that you will interact with as a data analytics professional are deeply rooted in the scientific method. As you continue in your data professional journey, embrace inquiry, intellectual discussion, and collaboration. Invite feedback and assess feedback. Remember, artificial intelligence still depends heavily on the instructions provided by data professionals. The more time and consideration that goes into the process of data analytics, the better the results.

Different industries have different standards. In your role as a data analytical professional, you will need to be aware of the standards for the industries you are working in. Each industry will have its own standards based on industry conventions.

Conventions that work well in the transportation industry may not necessarily be as high of a priority for the healthcare industry. For example, in transportation, data is collected to create predictive analytics models to analyze the best route based on traffic patterns. In the healthcare industry, data is analyzed in medical imaging, predicting genetic factors, and speeding up the development of treatments.

Data stewardship and ethics conversations

As you progress through your career as a data analytics professional, you will need to consider issues of ethical concern. For example, you may encounter situations where you address questions of bias or need to protect user data and personally identifiable information (PII). When these types of questions arise, many seek guidance and support from online communities of data professionals who have dealt with similar issues. The following graphics present scenarios involving these kinds of issues. You can also find text alternative versions of these conversations in the [Data stewardship and ethics conversations transcript](#).



Key takeaways

Data stewardship is the responsibility of every data professional. This responsibility goes beyond interactions with the data. By conducting your work in ways that are socially beneficial and inclusive, you will increase your ability to identify human bias. Guide your efforts through scientific and ethical principles and stay aware of possible bias throughout the data analysis process.